

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)	Quarks and leptons and mesons – all required	1			1		
		(ii)	Quarks and mesons	1			1		
	(b)	(i)	Anti-[electron] neutrino Accept: [electron] anti-neutrino		1		1		
		(ii)	Lepton number: $0 = 0 + 1 - 1$ must be in this order or described: lepton number is 0 on LHS. Electron has lepton number of +1, anti-electron neutrino has a lepton number of -1. [1] Accept $4 = 4 + 1 - 1$ must be in this order. Charge: B has one more proton than Be, but appearance of e^- means that charge is conserved or $+4 = +5 - 1 [+0]$ must be in this order Or: $0 \rightarrow 1 - 1 + 0$ must be in this order or $n = p + e^- + \text{anti-neutrino}$ so $d = u + e^- + \text{anti-neutrino}$ so $-\frac{1}{3} = \frac{2}{3} - 1 + 0$ [1]		2		2		
	(c)	(i)	$^{10}_4\text{Be}$ has 6 neutrons and $^{10}_5\text{B}$ has 5 neutrons	1			1		
		(ii)	LHS: udd [1] RHS: uud [1] N.B. bod if $udd \rightarrow uud + e^- + \text{anti-neutrino}$ seen Award 1 mark if above seen with quarks assigned to either the e^- and/or anti-neutrino Award 1 mark for $d \rightarrow u$ or d in $n \rightarrow u$ in p or similar Award 1 mark if equation written the wrong way around		2		2		
			Question 1 total	3	5	0	8	0	0